

FINAL EXPLANATION: BIOLOGY GRADE 11

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: BIOLOGY GRADE 11: GROWTH AND DEVELOPMENT IN ANIMALS (SUB-STRAND 3.2)

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

Driving Question: How does an animal outgrow its own hard shell, and why do some animals radically change shape while others just grow bigger? Think back to the empty, split skin (exuvia) on the maize leaf in the school garden, lying beside a larger newly emerged grasshopper or cicada. Use what you observed and learned in Lessons 1 to 5 to answer each section below. Write in full sentences, use correct scientific words (for example exoskeleton, moulting/ecdysis, metamorphosis, Brain Hormone, Ecdysone, Juvenile Hormone), and where asked, give a Claim, Evidence and Reasoning. You may draw and label diagrams or sketch graphs in the space under each prompt. Only describe patterns you actually saw in our class investigations.

Part 1: Outgrowing the Shell (Stepwise Growth)

Look back at the exuvia on the maize leaf beside the larger grasshopper or cicada. (a) Explain how the insect became bigger than the hard skin it was wearing. In your answer, say what the exoskeleton can and cannot do, and when in the moulting process the insect actually increases in size. (b) Describe the shape of the graph of nymph length against time that we plotted in class, and explain why it looks like that. Use the words moulting (ecdysis) and exuvia.

(a) The insect's hard exoskeleton cannot stretch, so the insect cannot grow smoothly inside it. Before moulting, the insect makes a new, larger cuticle underneath the old one. The old exoskeleton then splits down the back (ecdysis/moulting) and the insect pulls out, leaving the empty exuvia on the maize leaf. While the new cuticle is still soft, the insect swells and expands its body, and only then does the new cuticle harden. This is why the grasshopper or cicada is clearly bigger than the exuvia it left behind. It did not peel like sunburned skin; it replaced its whole shell. (b) The graph of nymph length against time was not a smooth line. It was a staircase: flat sections with no change in length while the hard shell was in place, and sudden jumps at each moult when the insect expanded in the soft new cuticle. This is intermittent (stepwise) growth, and it happens because the exoskeleton cannot stretch and must be shed before the insect can get bigger.

Part 2: Two Growth Patterns Compared

In class we plotted data for a Kenyan broiler chick, a child, and a grasshopper nymph. (a) Describe the shape of the growth curve for the chick and

(a) The chick and child graphs were smooth S-shaped (sigmoid) curves. The four phases are the lag phase (slow growth at first), the exponential phase (rapid growth), the deceleration phase (growth slowing down) and the stationary phase (size levels off at adult size). (b) Growth rate = (final value - initial value) / time interval. Growth rate is highest in the exponential phase and close to zero in the stationary phase. (c) The chick has no rigid shell, so its cells can divide and increase in size continuously, giving a smooth curve. It starts slowly, grows fast, then slows as it nears the adult size set by its genes and resources. The grasshopper

child, and name the four phases you identified. (b) State the formula for growth rate that we used and say in which phase growth rate is highest and in which phase it is close to zero. (c) Explain why the chick's graph looks different from the grasshopper nymph's graph.	nymph has a hard exoskeleton that cannot stretch, so its length stays flat and then jumps at each moult, giving a staircase (intermittent) curve.
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Part 3: The Hormone Control System

Moulting and metamorphosis are not random. Using the experiments on larvae that you analysed (removing the brain/neurosecretory cells, injecting Ecdysone, adding extra Juvenile Hormone, removing the corpora allata early), write a CER statement answering: What controls when an insect moults and what kind of moult it has? Include the role of Brain Hormone, Ecdysone and Juvenile Hormone in your reasoning.	Claim: Moulting and the type of moult are controlled by hormones working together as a control system, not by chance. Brain Hormone signals the prothoracic glands to release Ecdysone, which triggers moulting, while Juvenile Hormone (JH) decides what kind of moult happens. Evidence: When the brain/neurosecretory cells were removed, the larva did not moult. When Ecdysone was injected, the larva moulted again. When extra JH was given, the larva moulted into another larva instead of a pupa. When the corpora allata (which make JH) were removed early, the larva formed a small pupa early. Reasoning: Without Brain Hormone there is no signal to the prothoracic glands, so no Ecdysone and no moult, which shows the brain starts the process. Ecdysone gives the moult itself. High JH keeps the larval form at each moult, low JH allows pupation, and no JH allows the adult to emerge. So the combined action of the three hormones regulates both when moulting happens and what form results.
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Part 4: Two Pathways of Metamorphosis

Compare a grasshopper and a butterfly (or housefly). (a) Name the stages in each life cycle and say which is incomplete and which is complete metamorphosis. (b) Using Juvenile Hormone, explain why grasshopper nymphs look like small adults but a caterpillar looks nothing like the adult butterfly. (c) Give TWO ways that having separate larval and adult stages might help an insect survive.	(a) The grasshopper has incomplete metamorphosis with 3 stages: egg, nymph, adult (no pupa). The butterfly and housefly have complete metamorphosis with 4 stages: egg, larva, pupa, adult. (b) A grasshopper nymph keeps JH present through its nymph moults until the last moult, so each moult only makes a bigger version of the same shape, and it becomes an adult without a pupa. In a caterpillar, JH stays high during larval moults but falls sharply before the pupa stage. The low JH allows Ecdysone to trigger pupation and the complete change of body shape into the adult butterfly. (c) Larvae and adults use different foods, so they do not compete with each other for resources. The pupa stage helps the insect survive harsh conditions, such as a dry season, while it is motionless. These controlled stages let the animal use resources well and survive.
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Part 5: Putting It All Together (Final CER)

Write a complete CER answer to the Driving Question: How does an animal outgrow its own hard shell, and why do some animals radically change shape while others just grow bigger? Use the exuvia on the maize leaf, a growth curve, and the hormones in your evidence. Then, in one or two sentences, suggest a new question you still have about this topic.

Claim: An animal with a hard exoskeleton outgrows its shell by moulting under hormone control, and whether it changes shape completely or just gets bigger depends on how Juvenile Hormone changes over its life. **Evidence:** The empty, split exuvia on the maize leaf lay beside a larger grasshopper or cicada. Our nymph graph rose in steps, with flat sections and sudden jumps at moults, while the chick and child graphs were smooth S-shaped curves. Hormone experiments showed that removing the brain cells stopped moulting, injecting Ecdysone caused moulting, extra JH produced another larva, and removing the corpora allata early produced an early small pupa. Grasshoppers go from nymph to adult with no pupa, but butterflies have a pupa stage. **Reasoning:** The exoskeleton cannot stretch, so Brain Hormone causes the prothoracic glands to release Ecdysone, which triggers a new, larger cuticle to form under the old one. The old one splits and is left as the exuvia, and the insect expands while the new cuticle is soft, so growth happens in steps (intermittent growth). Animals without a rigid shell, like vertebrates, can grow continuously, giving a sigmoid curve with lag, exponential, deceleration and stationary phases. In grasshoppers, JH stays present until the last moult, so the nymphs look like small adults (incomplete metamorphosis). In butterflies, JH falls before pupation, so the larva changes completely through a pupa into the adult (complete metamorphosis). **New question:** How might pesticides used by farmers disrupt these hormone pathways in agricultural pests?

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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Explaining stepwise growth and the exuvia	Clearly explains that the exoskeleton cannot stretch, a new larger cuticle forms beneath, the old one splits to leave the exuvia, and the insect expands while the new cuticle is soft. Links this accurately to the stepped graph.	Explains that the insect moults and gets bigger afterwards, and describes the stepped graph, but leaves out one detail such as the new cuticle forming first or the soft cuticle expanding.	States that the insect sheds its skin but does not explain why it must moult or how it becomes bigger, or treats the exuvia like peeling skin.
Comparing growth curves (sigmoid vs intermittent)	Correctly names all four phases of the S-curve, uses the growth rate formula, identifies where rate is highest and lowest, and explains the difference from the insect staircase using the rigid exoskeleton.	Describes both curve shapes and names most phases or the formula, with a reasonable but incomplete explanation of the difference.	Describes only one curve or confuses the shapes, with few or incorrect phases and no clear link to the exoskeleton.
Hormonal control (Brain Hormone, Ecdysone, JH)	Accurately explains the sequence from Brain Hormone to prothoracic glands to Ecdysone, and the role of high, falling and absent JH, using specific experimental evidence in a full CER.	Names the three hormones and their main roles and uses some experimental evidence, with minor gaps in the reasoning.	Names hormones but mixes up their roles, or gives claims with little or no evidence or reasoning.
Metamorphosis pathways and survival value	Gives correct stages of both pathways, explains with JH why shapes differ, and gives two clear survival advantages.	Gives correct stages and a mostly accurate explanation of the shape difference, with one survival advantage.	Lists stages with errors or does not link the difference to JH or survival.
Final CER and scientific	A complete, well-organised Claim,	A mostly complete CER that answers the	An incomplete CER that answers only

communication	Evidence and Reasoning that answers both parts of the Driving Question using class evidence, correct vocabulary, and a thoughtful new question.	Driving Question with some class evidence and mostly correct vocabulary.	part of the Driving Question, with little evidence and limited scientific vocabulary.
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